BAHIRDAR UNIVERSITY

BAHIRDAR INSTITUTION TECHNOLOGY

FUCALITY OF COMPUTING

DEPARTMENT OF IT BED

COURSE NAME :OPRATING SYSTEM PROJECT :

1.OPRATING SYSTEM :UBUNTU 22.04 LTS INSTALLATION

2.WORLD WIDE BRADE :ACER PREDATOR

3.WORLD WIDE MOBILE PHONE BRAND :MICROMAX

NAME :HABTAMU MEKET.... ... ID :1601651

Operating System: Ubuntu 22.04 LTS

Worldwide PC Brand: Acer Predator

Worldwide Mobile Phone Brand: Micromax

1. Introduction

Virtualizationnn allows users to run multiple operating systems on a single physical machine through software abstraction. This project focuses on installing Ubuntu 22.04 LTS (Jammy Jellyfish) in a virtual environment. Ubuntu 22.04 is a Long-Term Support release supported until April 2027, making it ideal for learning and development purposes. Virtualization provides a safe, isolated environment to practice OS installation without risking the host system.

Ubuntu 22.04 LTS is installed using virtualization tools to understand OS installation, configuration, and management in a safe environment. The project also studies Acer Predator as a worldwide PC brand and Micromax as a worldwide mobile phone brand to understand how operating systems interact with different hardware platforms.

2. Objectives

2.1 Objectives of Knowing Virtual Installation of Operating Systems

To understand OS installation procedures in a controlled virtual environment

To apply theoretical OS concepts such as boot process, file systems, and disk partitioning

To gain practical experience with operating systems without requiring extra hardware

To develop troubleshooting skills during OS installation and configuration

To experiment with OS features like memory allocation, CPU usage, and device drivers

To understand virtualization technology and its importance in modern computing

To prepare for real-world tasks in system administration and OS management

2.2 Objectives of Knowing Worldwide PC Brands (Acer Predator)

To identify global PC manufacturers capable of running modern operating systems

To understand how hardware specifications affect OS performance

To analyze OS compatibility with high-performance hardware

To appreciate industry standards in PC manufacturing

To make informed decisions when selecting hardware for OS installation

To understand system architectures such as Intel and AMD

Acer Predator represents high-performance PCs suitable for advanced OS tasks like virtualization and multitasking.

3. Operating System Selection

The operating system selected for this project is Ubuntu 22.04 LTS.

Reasons for selection:

Long Term Support (LTS)

Stable and secure

Widely used in education and industry

Strong support for virtualization tools

3. Requirements

Hardware (Virtual Machine):

· Processor: 2 cores minimum (x64)

· RAM: 8 GB minimum

· Storage: 25 GB minimum

· Graphics: SVGA compatible

Software:

· Host OS: Windows 10

· Virtualization Tool: Oracle VM VirtualBox 7.0

· Installation Media: ubuntu-22.04.4-desktop-amd64.iso

4.0 INSTALLATION STEPS

Phaseee 1: Preparation

Step 1: Enable Virtualization in BIOS/UEFI

1. Restart computer and press appropriate key (F2, F10, DEL, ESC)

2. Navigate to Advanced CPU Settings or Security Settings

3. Enable:

   · Intel Virtualization Technology (VT-x)

   · VT-d (if available)

4. Save changes and exit

Step 2: Download Required Files

· Download Oracle VM VirtualBox from official website

· Download Ubuntu 22.04 LTS ISO (approx. 3.9GB)

· Verify ISO checksum for integrity

Step 3: Install VirtualBox

1. Run VirtualBox installer as administrator

2. Accept license agreement

3. Choose installation location (default recommended)

4. Select components (all selected by default)

5. Complete installation and restart if prompted

Phase 2: Virtual Machine Creation

Step 4: Create New Virtual Machine

Machine Name:

Habtamu Ubuntu_22.04

Type: Linux

Version: Ubuntu (64-bit)

Memory Size: 4096 MB (4GB)

Hard Disk: Create virtual hard disk now

File Type: VDI (VirtualBox Disk Image)

Storage: Dynamically allocated

Size: 25.00 GB

Step 5: Configure VM Settings

1. System → Processor: 2 CPUs

2. Display → Video Memory: 128 MB,

Enable 3D Acceleration

3. Storage: Attach Ubuntu ISO to optical drive

4. Network: NAT (Network Address Translation)

Phase 3: Ubuntu Installation

Step 6: Start Installation

1. Start the virtual machine

2. Select "Try or Install Ubuntu"

3. Choose language and keyboard layout

4. Select "Normal Installation" with:

   · [√] Download updates while installing

   · [√] Install third-party software

Step 7: Disk Partitioning

· Select "Erase disk and install Ubuntu"

· This creates:

  · EFI System Partition: 512MB

  · Root Partition: Remaining space (ext4 filesystem)

· No separate swap partition (uses swap file)

Step 8: User Configuration

User Details:

- Your name: habtamu ubuntu22.04

- Your computer's name: habtamu ubuntu 22.04

- Pick a username: habtamu ubuntu22.04

- Choose a password: 1601651

- Confirm password: 1601651

- [√] Require my password to log in

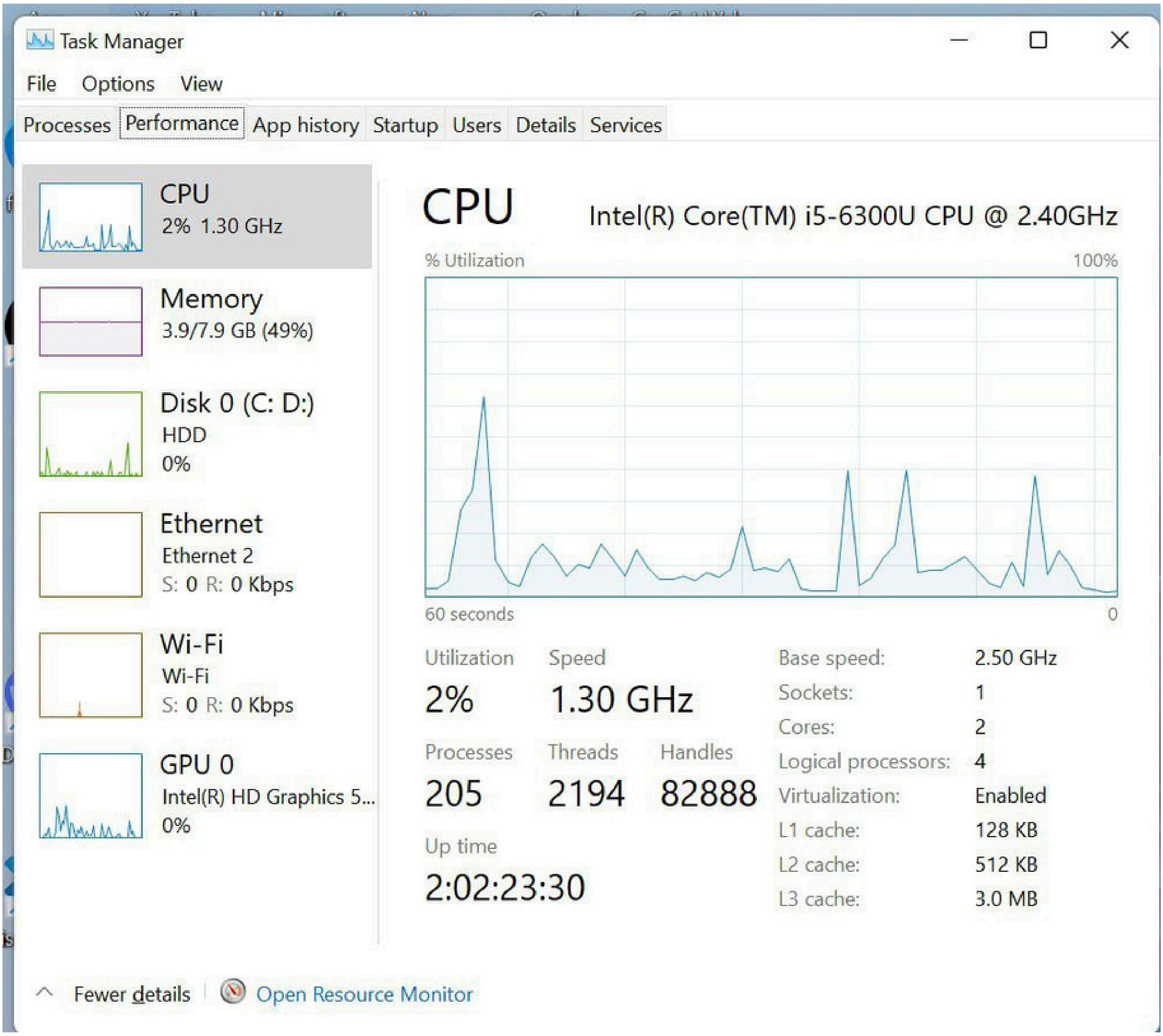
Step 9: Complete Installation

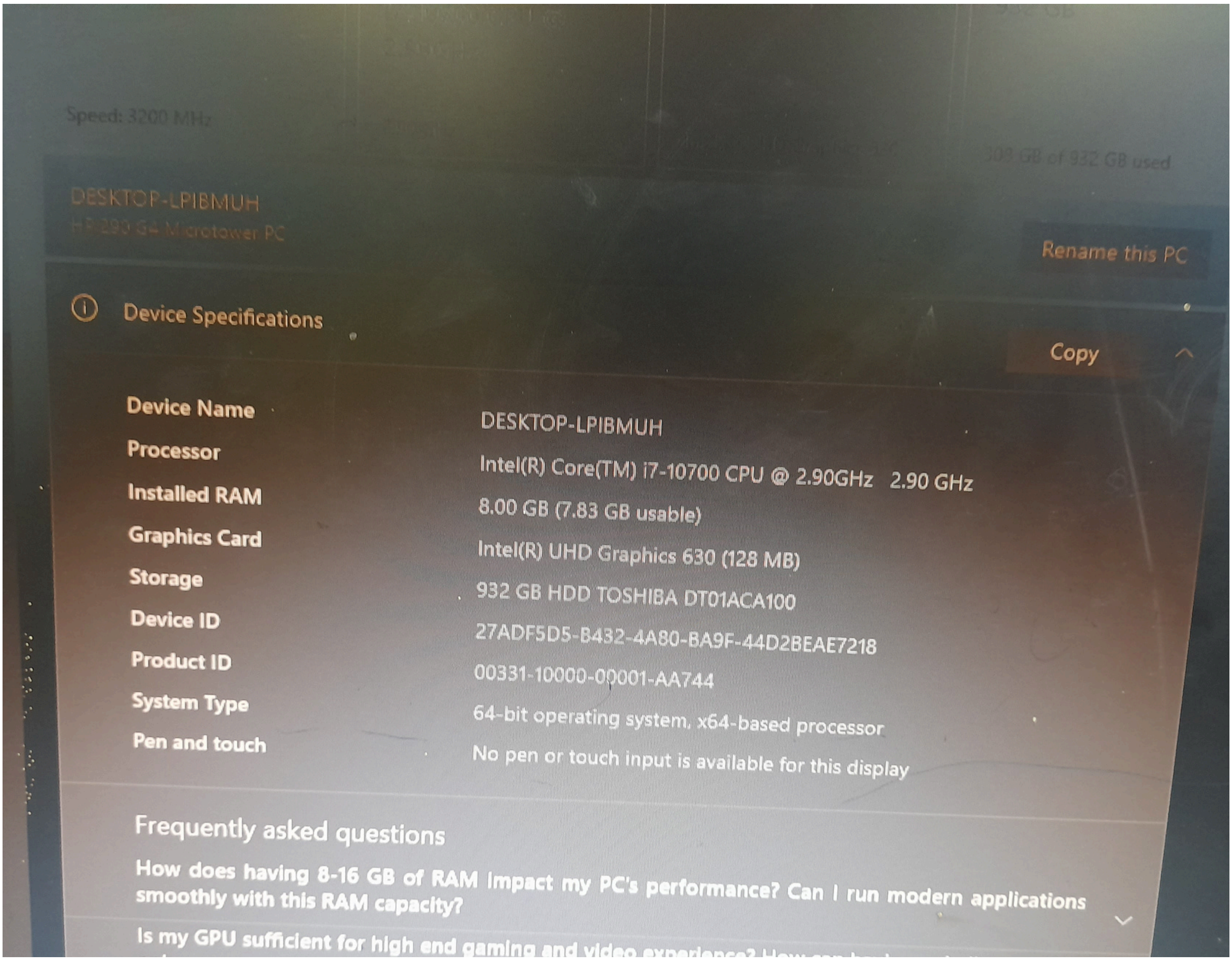
· Wait for installation to complete (20-30 minutes)

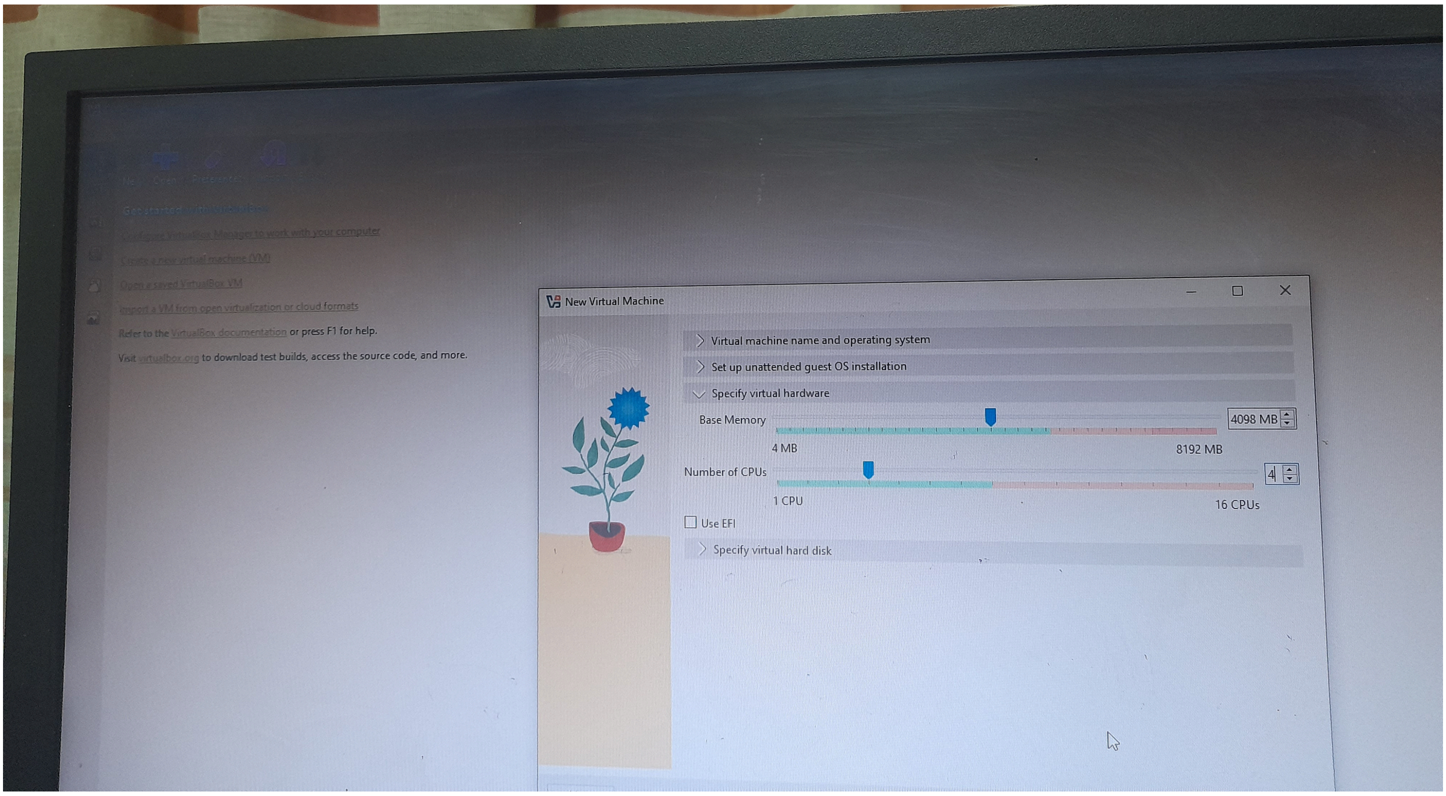
· Remove installation media when prompted

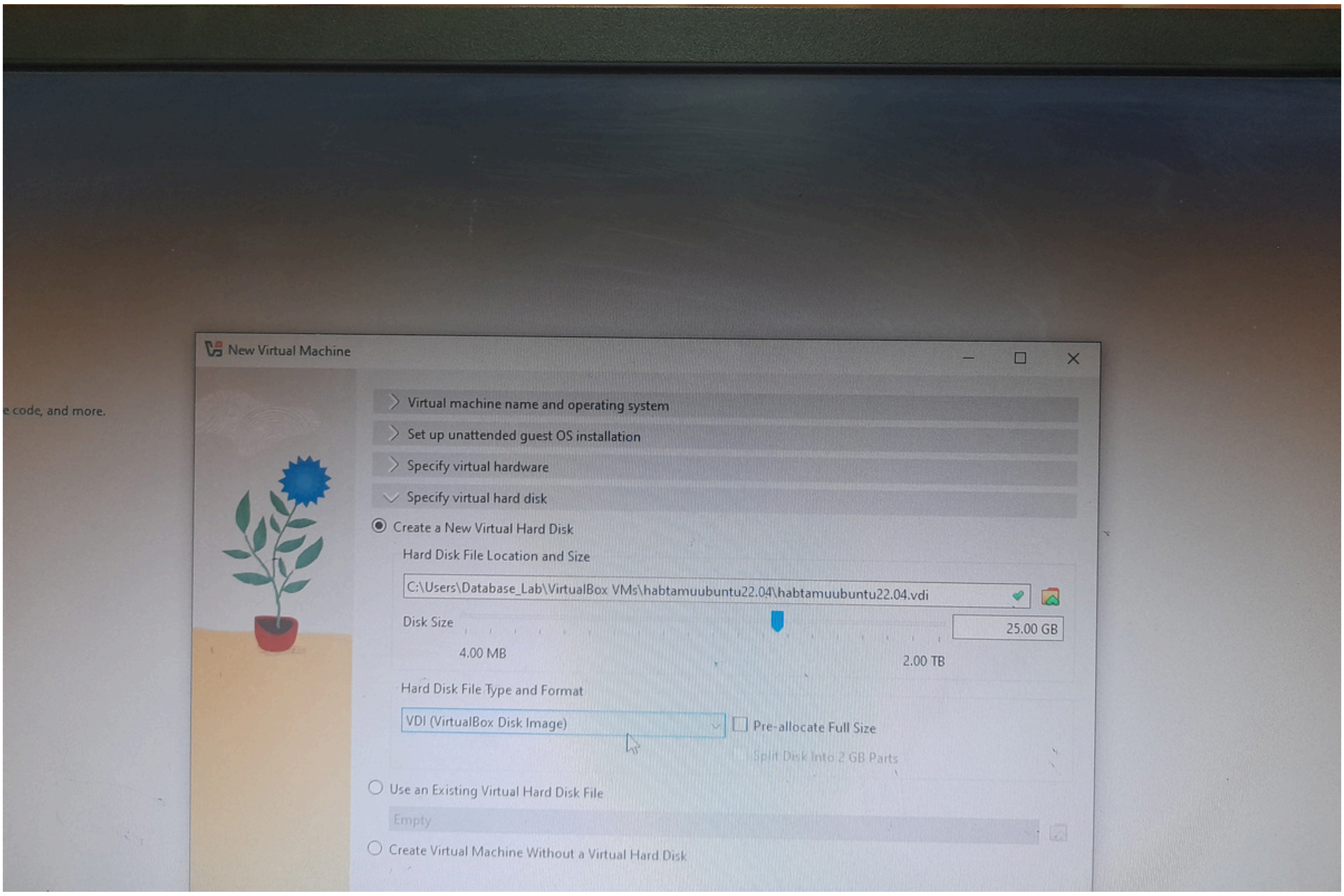
· Click "Restart Now"

.Complete installation and restart

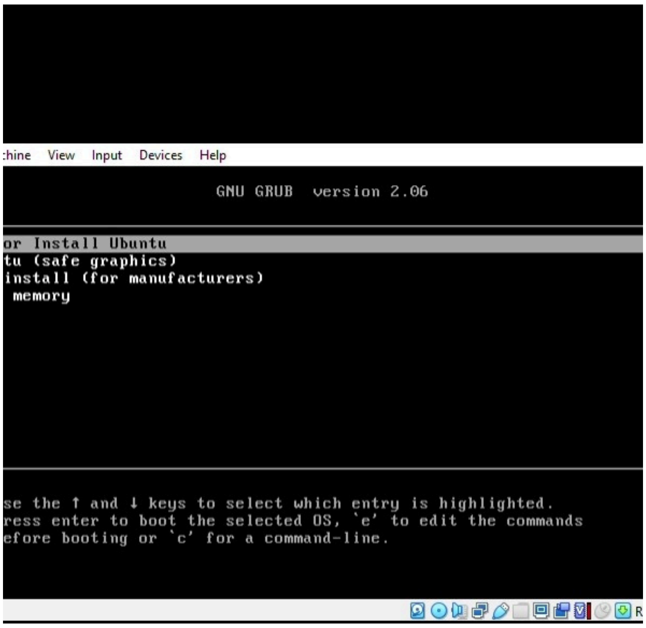

Task Manager
File
Options
View
Processes
Performance
App history
Startup
Users
Details
Services
CPU
2% 1.30 GHz
Memory
3.9/7.9 GB (49%)
Disk 0 (C: D:)
HDD
0%
Ethernet
Ethernet 2
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 0 Kbps
GPU 0
Intel(R) HD Graphics 5...
0%
CPU
Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz
% Utilization
100%
60 seconds
0
Utilization
2%
Speed
1.30 GHz
Processes
205
Threads
2194
Handles
82888
Up time
2:02:23:30
Base speed: 2.50 GHz
Sockets: 1
Cores: 2
Logical processors: 4
Virtualization: Enabled
L1 cache: 128 KB
L2 cache: 512 KB
L3 cache: 3.0 MB
Fewer details
Open Resource Monitor


DESKTOP-LPIBMUH
Rename this PC
Device Specifications
Copy
Device Name
DESKTOP-LPIBMUH
Processor
Intel(R) Core(TM) i7-10700 CPU @ 2.90GHz 2.90 GHz
Installed RAM
8.00 GB (7.83 GB usable)
Graphics Card
Intel(R) UHD Graphics 630 (128 MB)
Storage
932 GB HDD TOSHIBA DT01ACA100
Device ID
27ADF5D5-B432-4A80-BA9F-44D2BEAE7218
Product ID
00331-10000-00001-AA744
System Type
64-bit operating system, x64-based processor
Pen and touch
No pen or touch input is available for this display
Frequently asked questions
How does having 8-16 GB of RAM impact my PC's performance? Can I run modern applications smoothly with this RAM capacity?

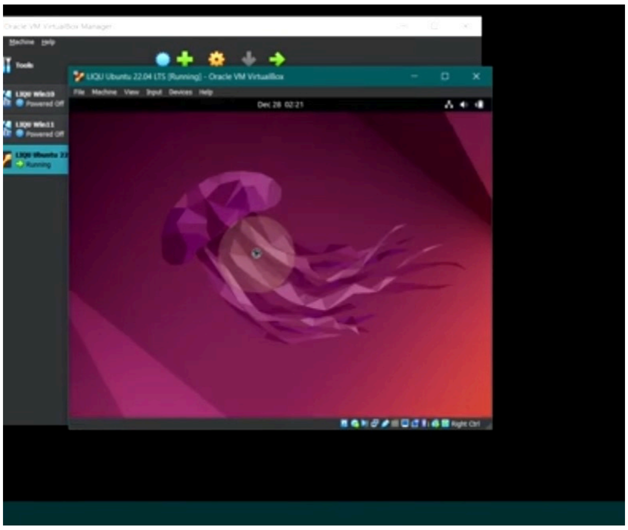

New Virtual Machine
Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Base Memory
4098 MB
4 MB
8192 MB
Number of CPUs
4
1 CPU
16 CPUs
Use EFI
Specify virtual hard disk

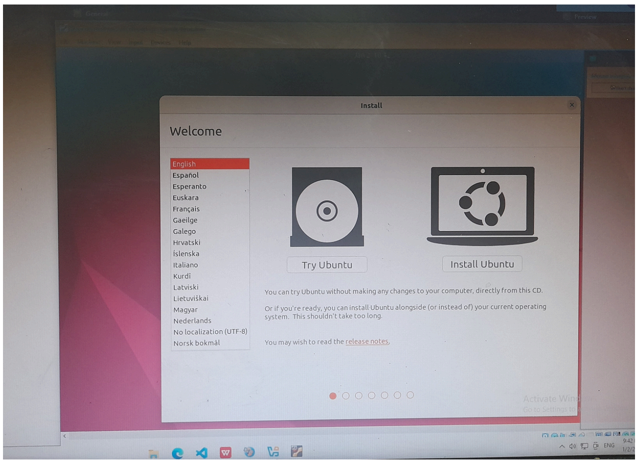


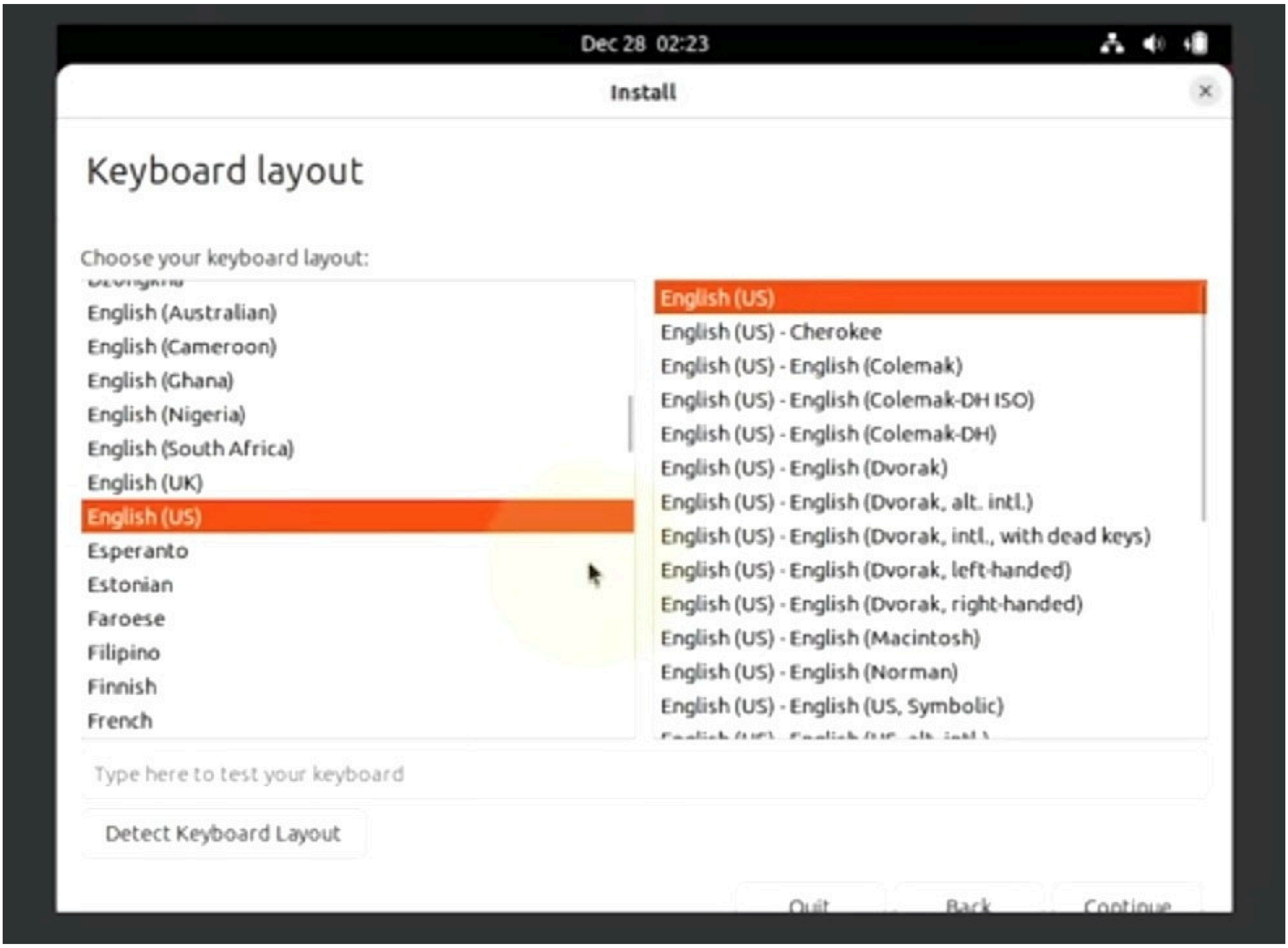
New Virtual Machine
Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Create a New Virtual Hard Disk
Hard Disk File Location and Size
C:\Users\Database_Lab\VirtualBox VMs\habtamuubuntu22.04\habtamuubuntu22.04.vdi
Disk Size
25.00 GB
4.00 MB
2.00 TB
Hard Disk File Type and Format
VDI (VirtualBox Disk Image)
Pre-allocate Full Size
Split Disk Into 2 GB Parts
Use an Existing Virtual Hard Disk File
Empty
Create Virtual Machine Without a Virtual Hard Disk

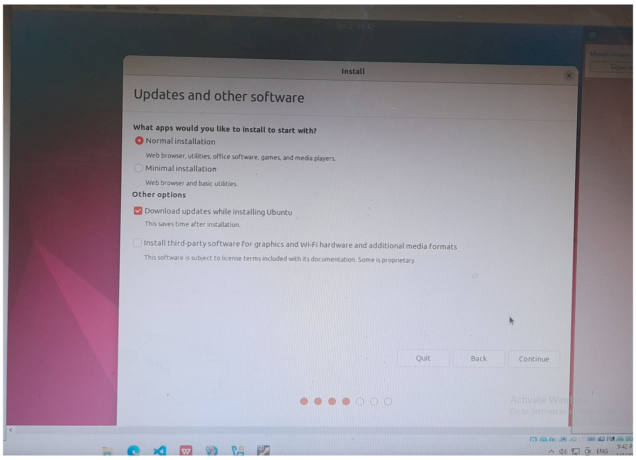


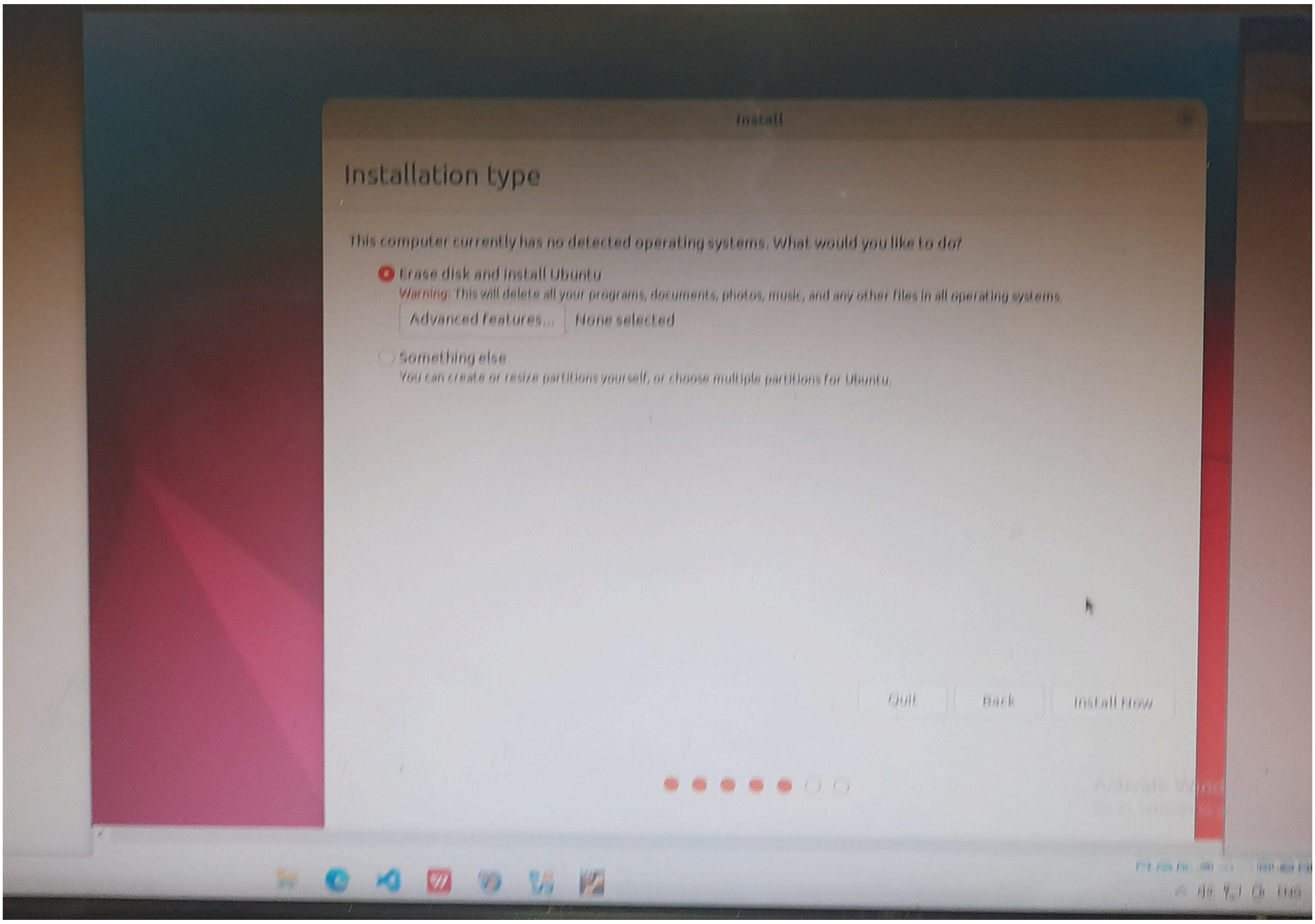
View Input Devices Help
GNU GRUB version 2.06
or Install Ubuntu
tu (safe graphics)
install (for manufacturers)
memory
se the ↑ and ↓ keys to select which entry is highlighted.
ress enter to boot the selected OS, `e' to edit the commands
efore booting or `c' for a command-line.

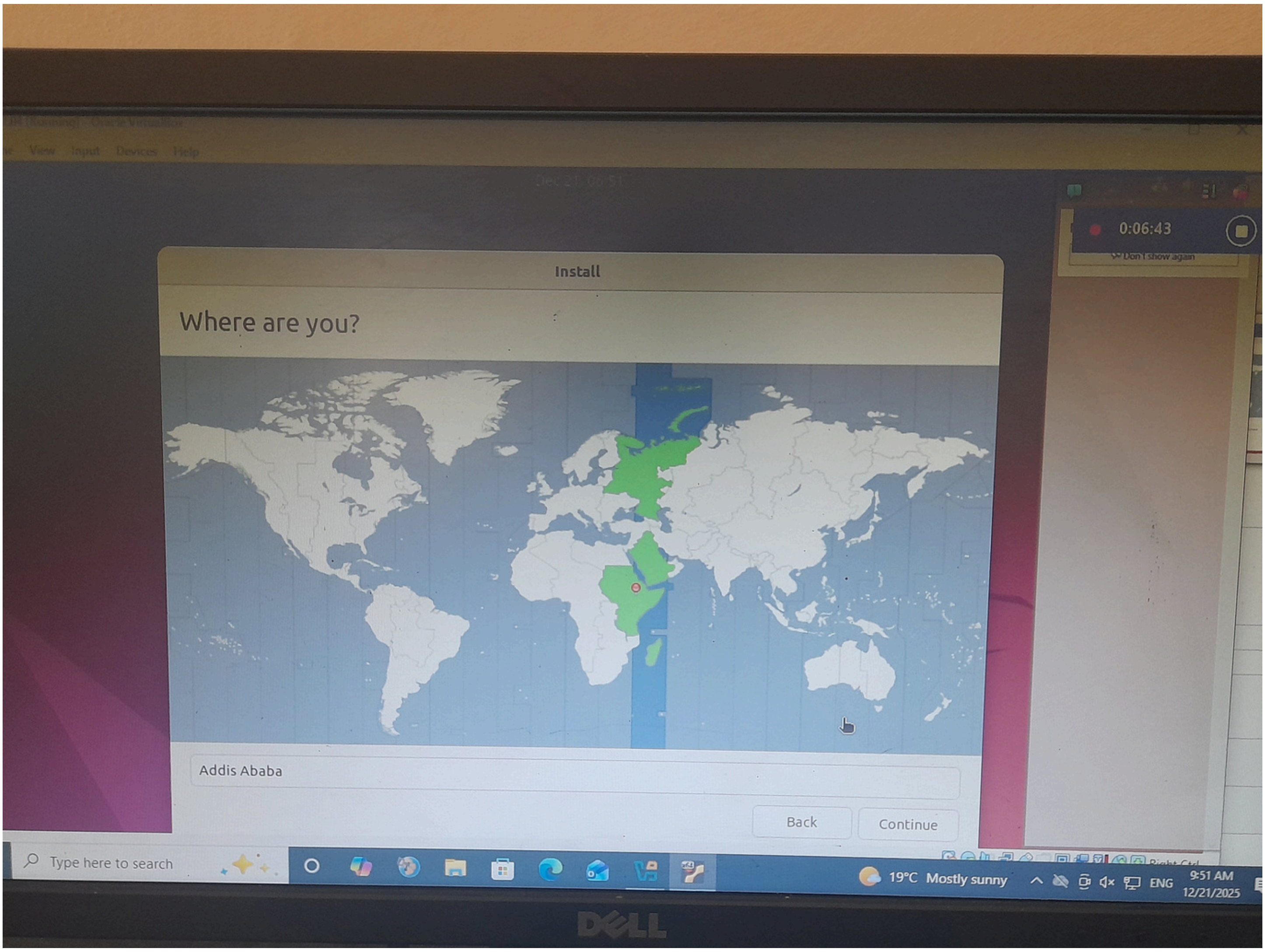

Dec 28 02:21

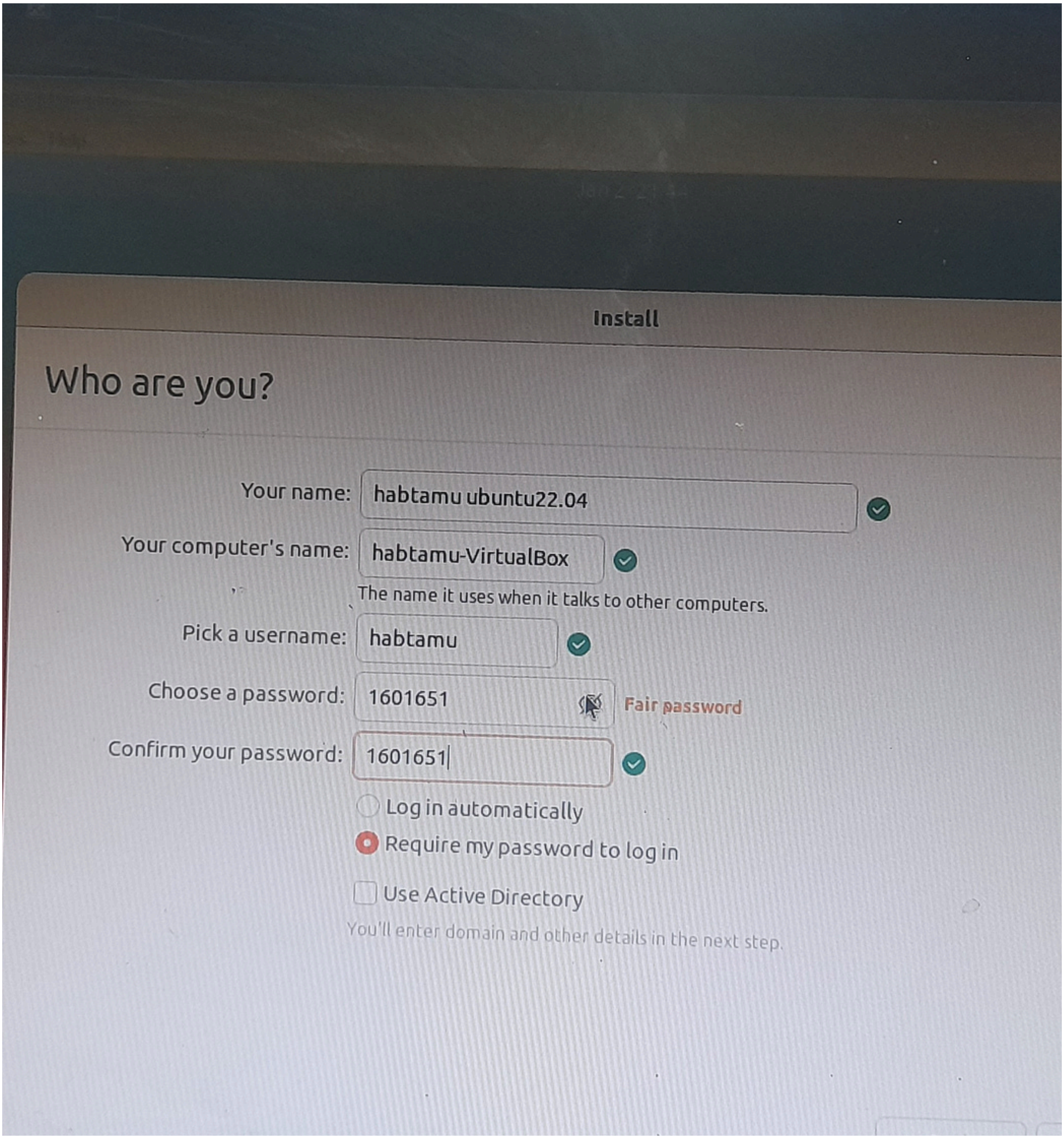

Install
Welcome
English
Español
Esperanto
Euskara
Français
Gaeilge
Galego
Hrvatski
Íslenska
Italiano
Kurdî
Latviski
Lietuviškai
Magyar
Nederlands
No localization (UTF-8)
Norsk bokmål
Try Ubuntu
Install Ubuntu
You can try Ubuntu without making any changes to your computer, directly from this CD.
Or if you're ready, you can install Ubuntu alongside (or instead of) your current operating system. This shouldn't take too long.
You may wish to read the release notes.

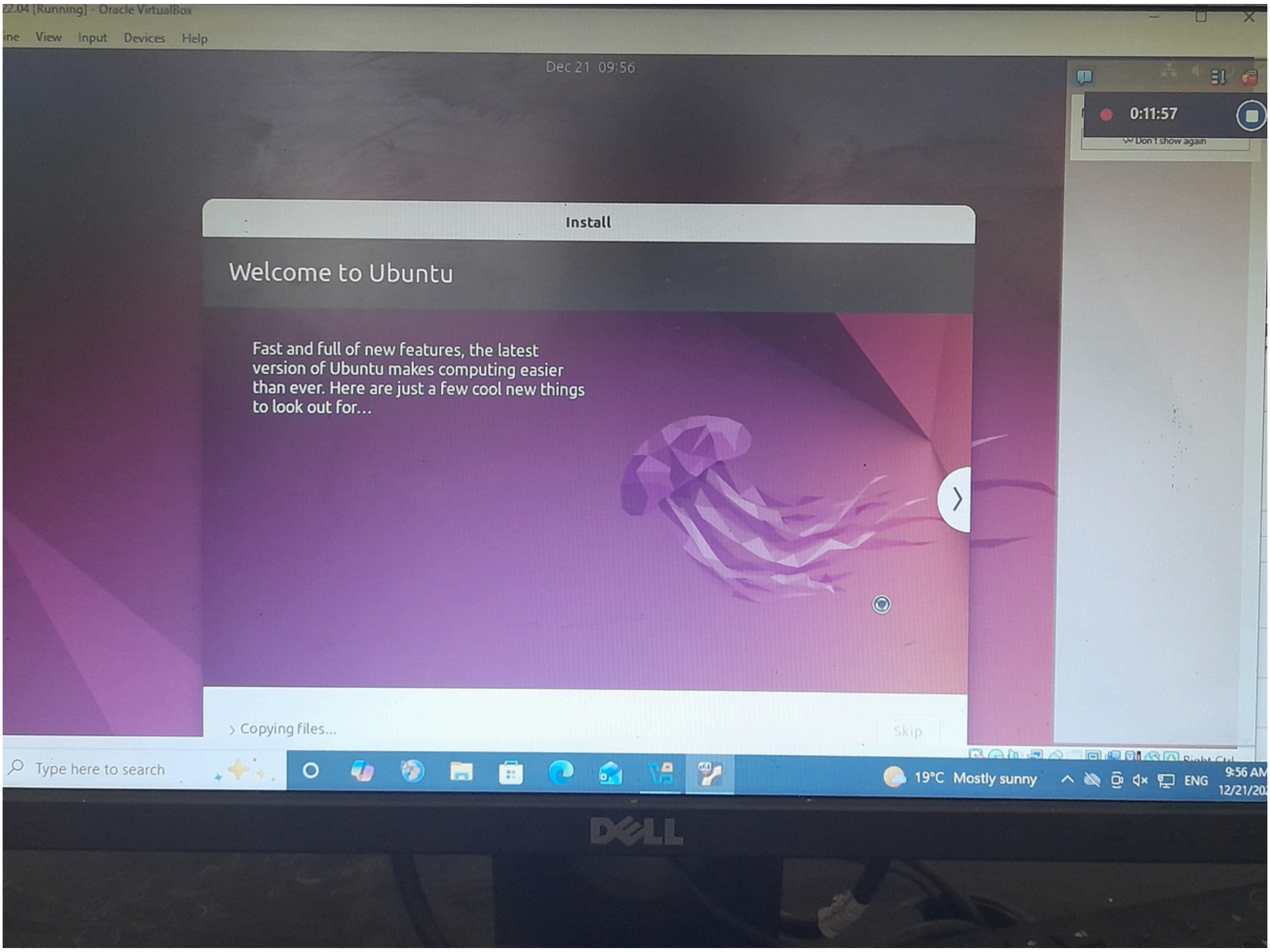

Dec 28 02:23
Install
Keyboard layout
Choose your keyboard layout:
English (Australian)
English (Cameroon)
English (Ghana)
English (Nigeria)
English (South Africa)
English (UK)
English (US)
Esperanto
Estonian
Faroese
Filipino
Finnish
French
English (US)
English (US) - Cherokee
English (US) - English (Colemak)
English (US) - English (Colemak-DH ISO)
English (US) - English (Colemak-DH)
English (US) - English (Dvorak)
English (US) - English (Dvorak, alt. intl.)
English (US) - English (Dvorak, intl., with dead keys)
English (US) - English (Dvorak, left-handed)
English (US) - English (Dvorak, right-handed)
English (US) - English (Macintosh)
English (US) - English (Norman)
English (US) - English (US, Symbolic)
Type here to test your keyboard
Detect Keyboard Layout

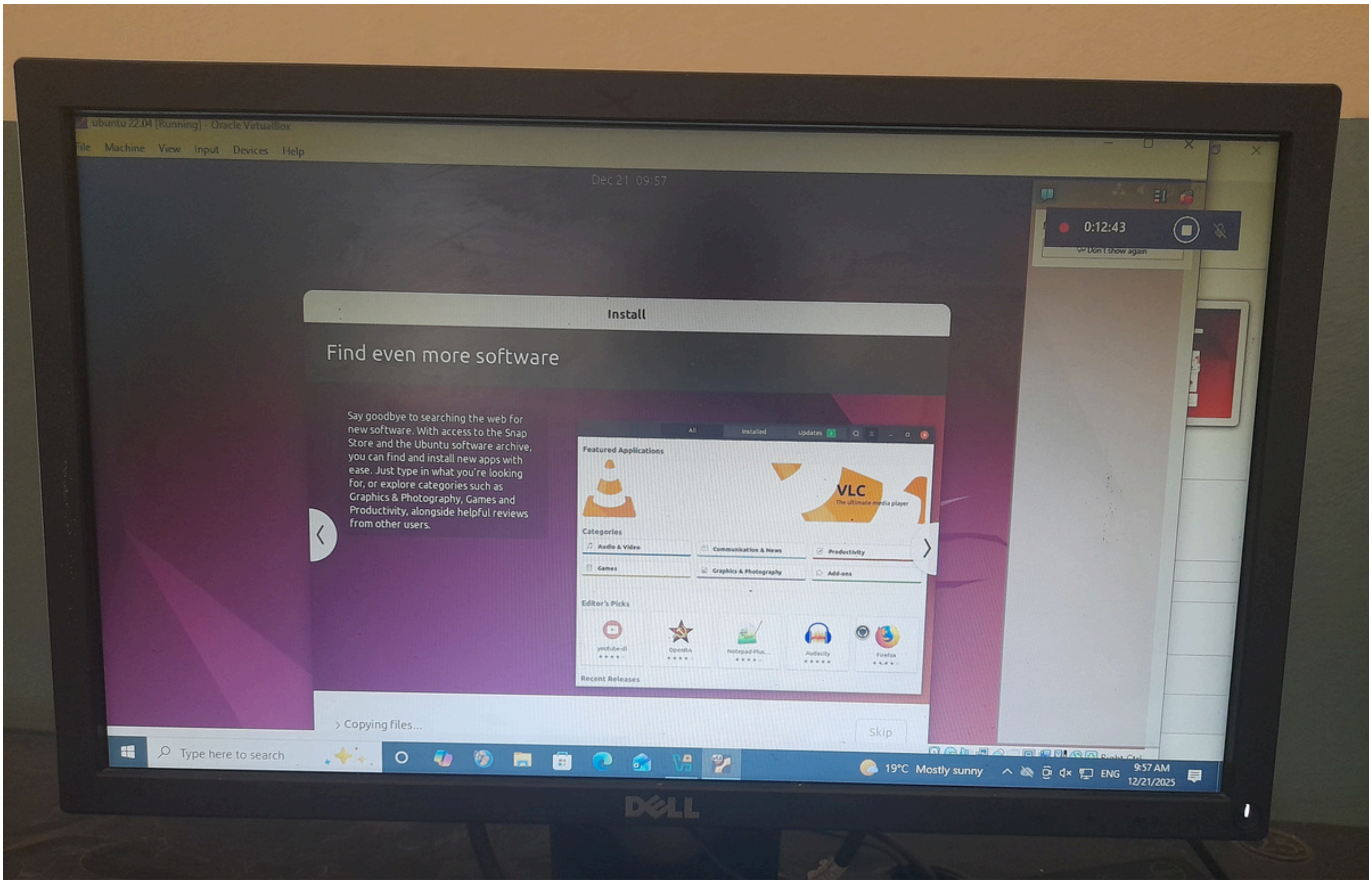

Install
Updates and other software
What apps would you like to install to start with?
Normal installation
Web browser, utilities, office software, games, and media players.
Minimal installation
Web browser and basic utilities.
Other options
Download updates while installing Ubuntu
This saves time after installation.
Install third-party software for graphics and Wi-Fi hardware and additional media formats
This software is subject to license terms included with its documentation. Some is proprietary.
Quit
Back
Continue

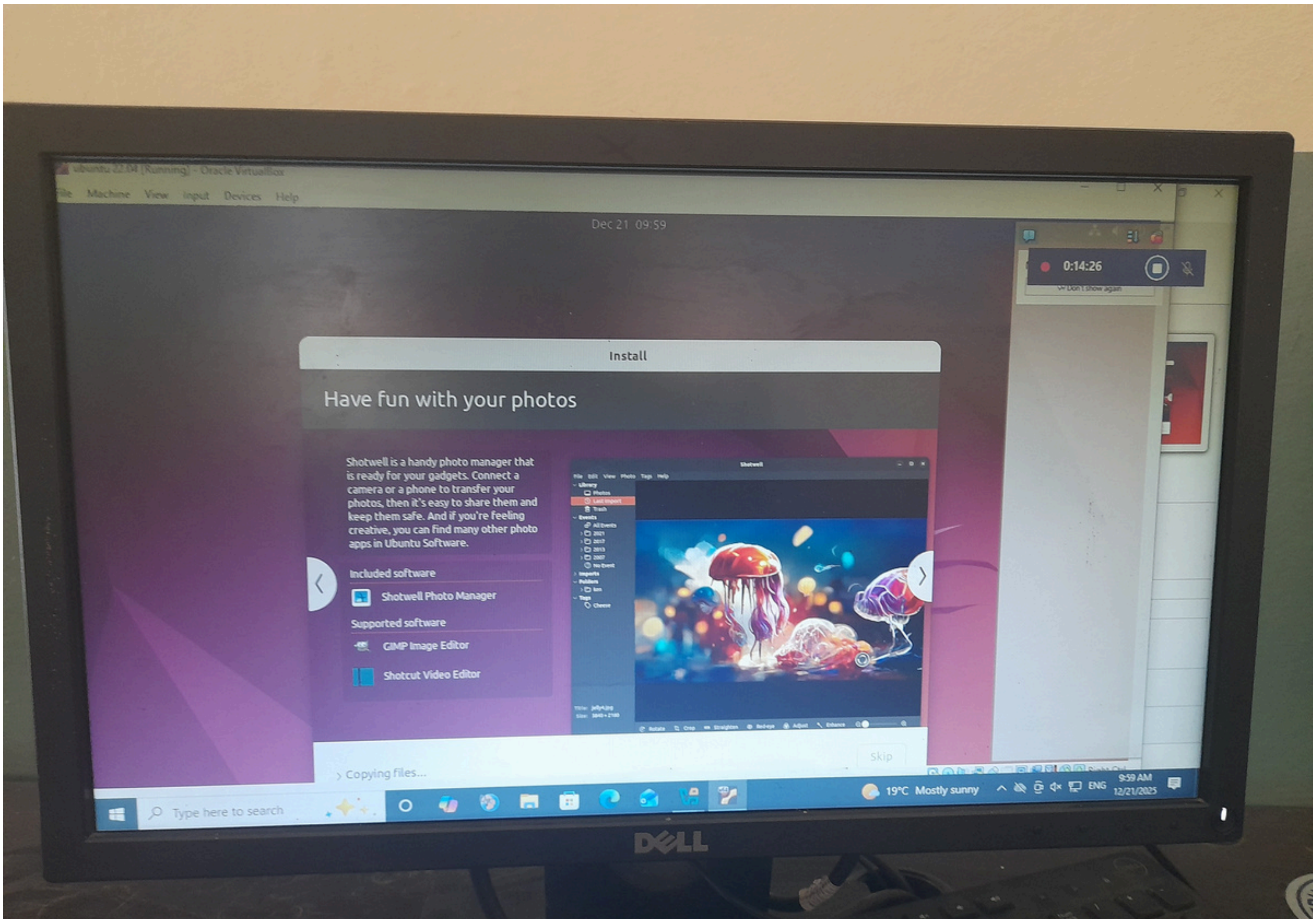

Install
Installation type
This computer currently has no detected operating systems. What would you like to do?
Erase disk and install Ubuntu
Warning: This will delete all your programs, documents, photos, music, and any other files in all operating systems.
Advanced features...
None selected
Something else
You can create or resize partitions yourself, or choose multiple partitions for Ubuntu.
Quit
Back
Install Now

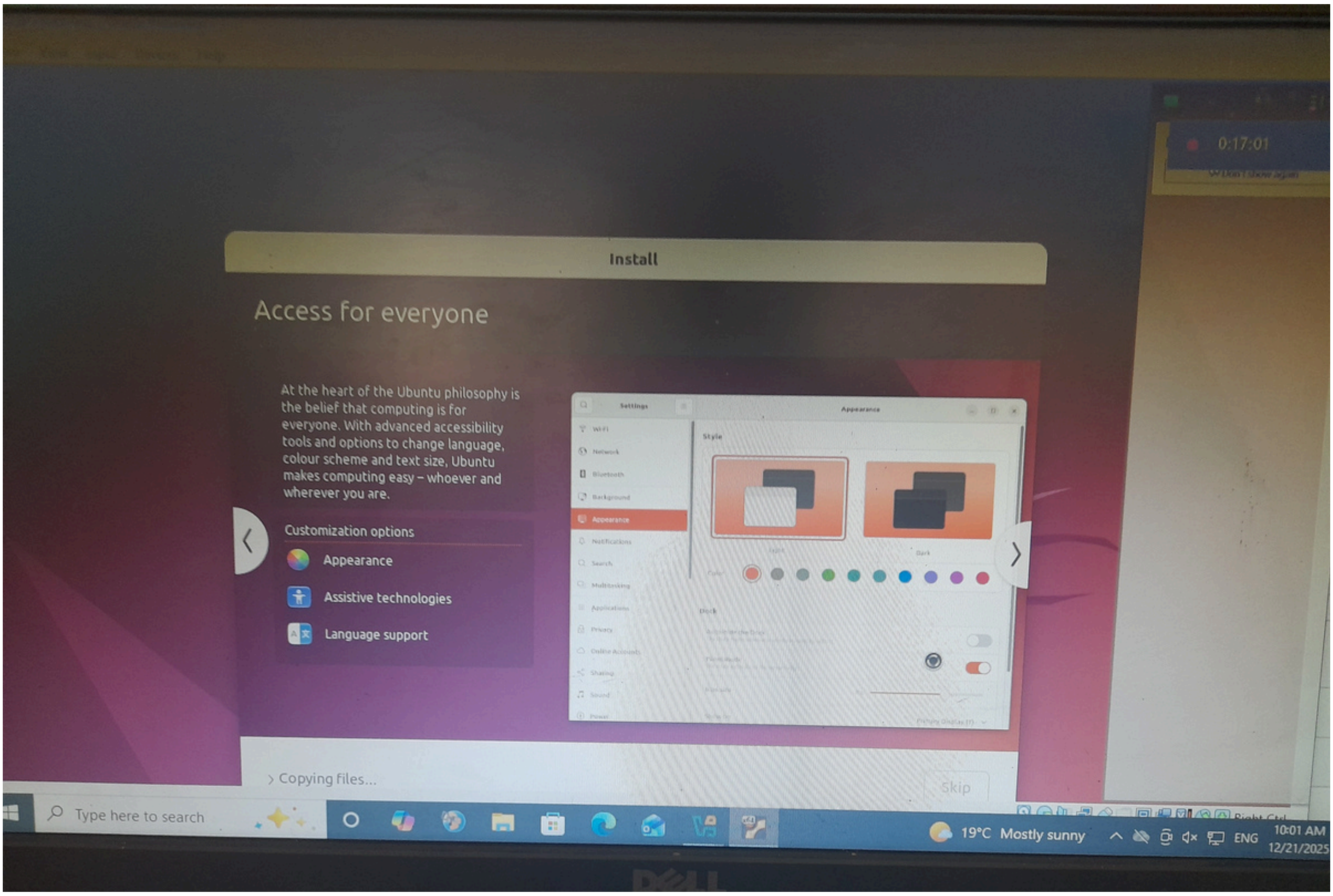

0:06:43
Install
Where are you?
Addis Ababa
Back
Continue
Type here to search
19°C Mostly sunny
ENG
9:51 AM
12/21/2025

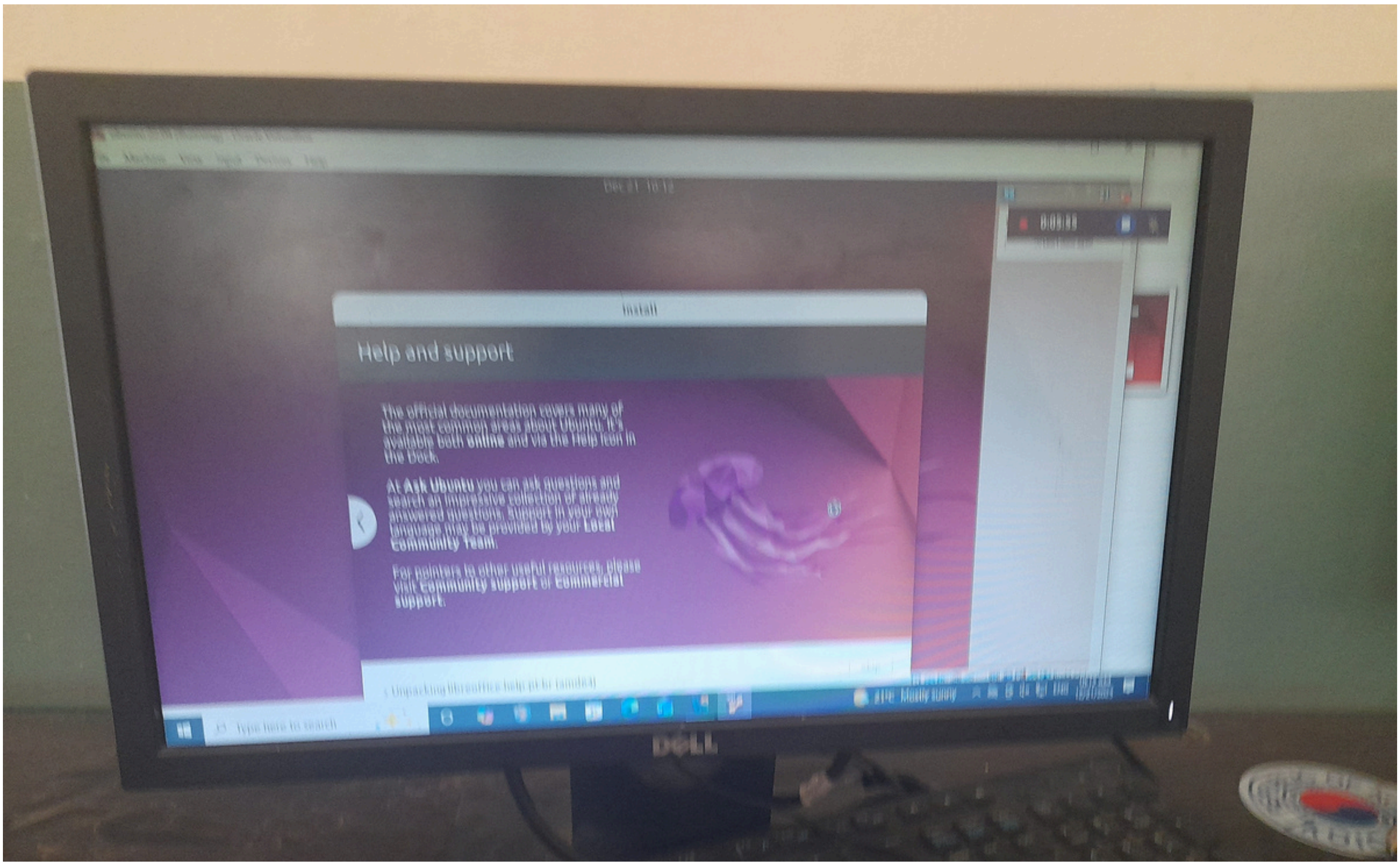

Install
Who are you?
Your name:
habtamu ubuntu22.04
Your computer's name:
habtamu-VirtualBox
The name it uses when it talks to other computers.
Pick a username:
habtamu
Choose a password:
1601651
Fair password
Confirm your password:
1601651
Log in automatically
Require my password to log in
Use Active Directory
You'll enter domain and other details in the next step.

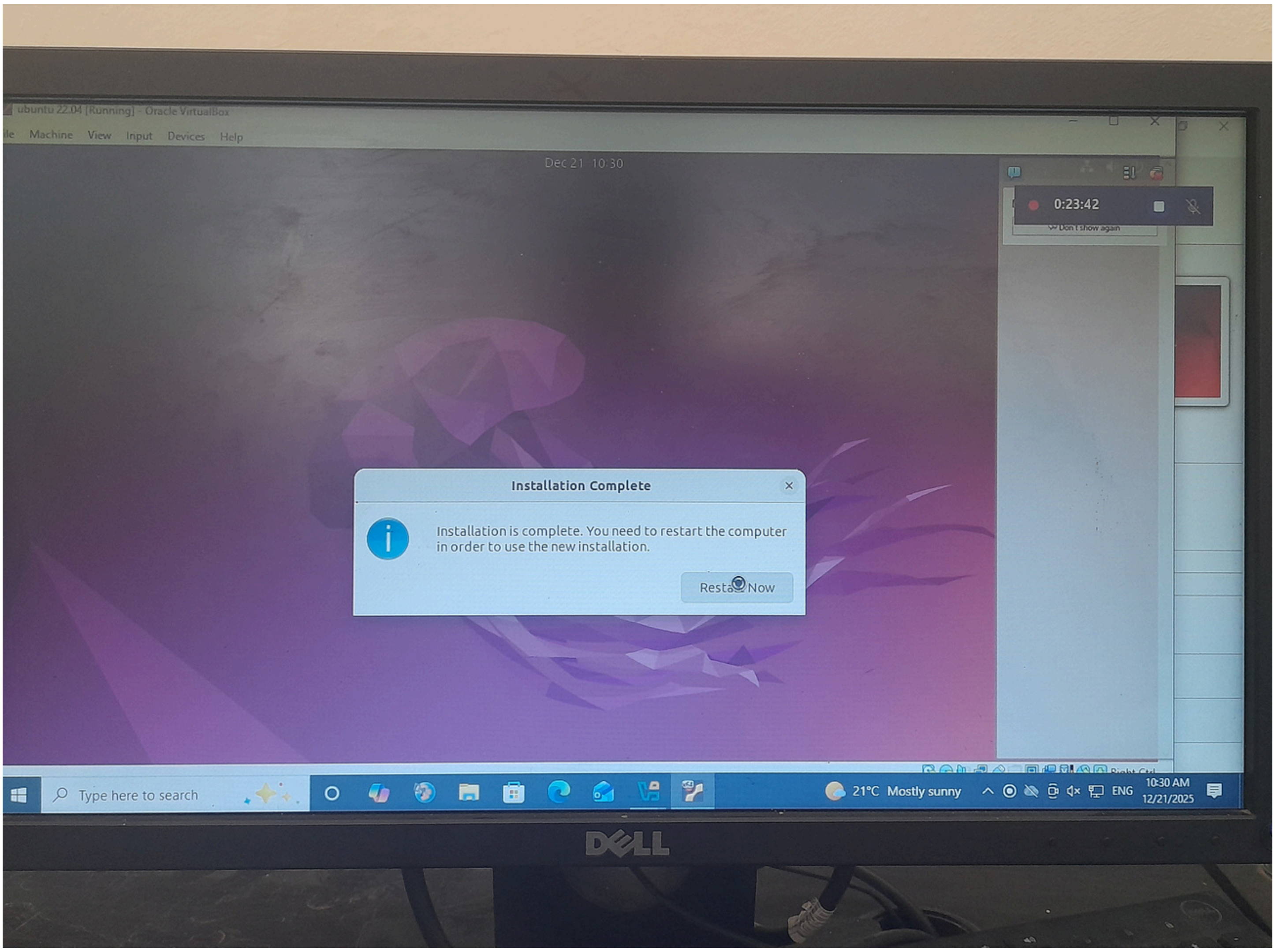

22.04 [Running] - Oracle VirtualBox
View Input Devices Help
Dec 21 09:56
0:11:57
Install
Welcome to Ubuntu
Fast and full of new features, the latest version of Ubuntu makes computing easier than ever. Here are just a few cool new things to look out for...
Copying files...
Skip
Type here to search
19°C Mostly sunny
ENG
9:56 AM

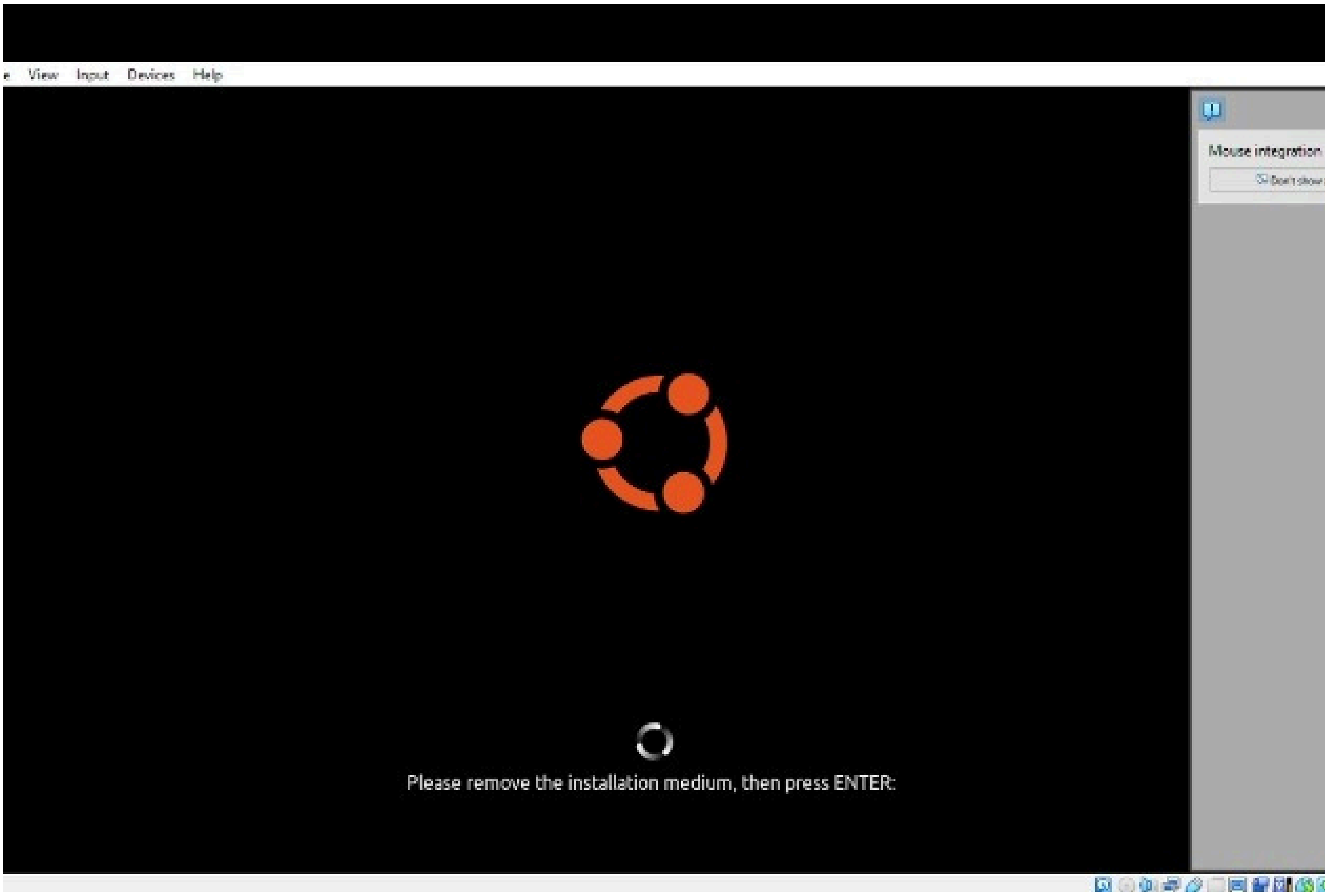

Dec 21 09:57
0:12:43
Install
Find even more software
Say goodbye to searching the web for new software. With access to the Snap Store and the Ubuntu software archive, you can find and install new apps with ease. Just type in what you're looking for, or explore categories such as Graphics & Photography, Games and Productivity, alongside helpful reviews from other users.
Copying files...
Skip
Type here to search
19°C Mostly sunny
ENG
9:57 AM
12/21/2025
DELL

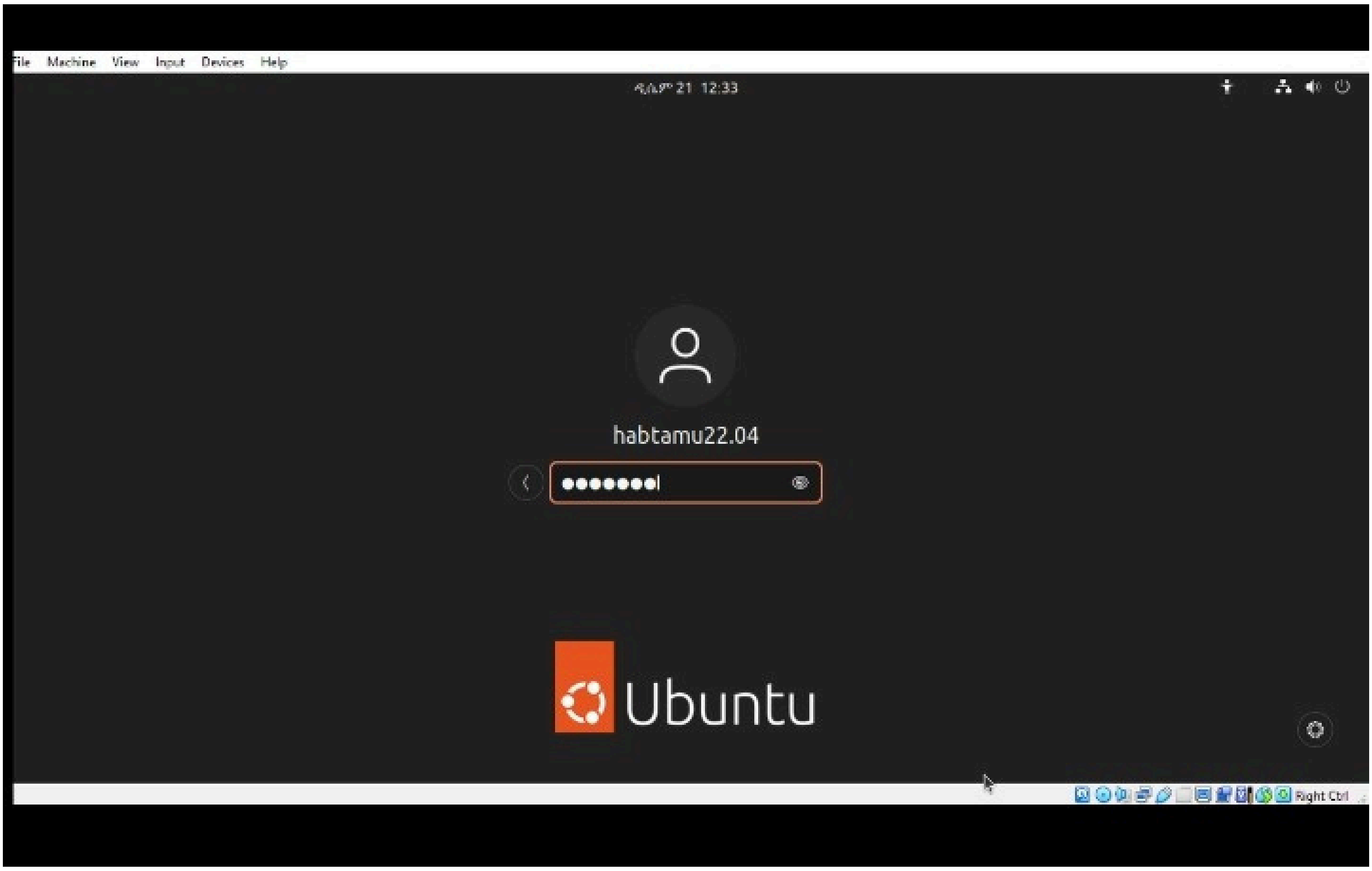

Install
Have fun with your photos
Shotwell is a handy photo manager that is ready for your gadgets. Connect a camera or a phone to transfer your photos, then it's easy to share them and keep them safe. And if you're feeling creative, you can find many other photo apps in Ubuntu Software.
Included software
Shotwell Photo Manager
Supported software
GIMP Image Editor
Shotcut Video Editor
Copying files...
Skip
Dec 21 09:59
0:14:26
Type here to search
19°C Mostly sunny
ENG
9:59 AM
12/21/2025
DELL

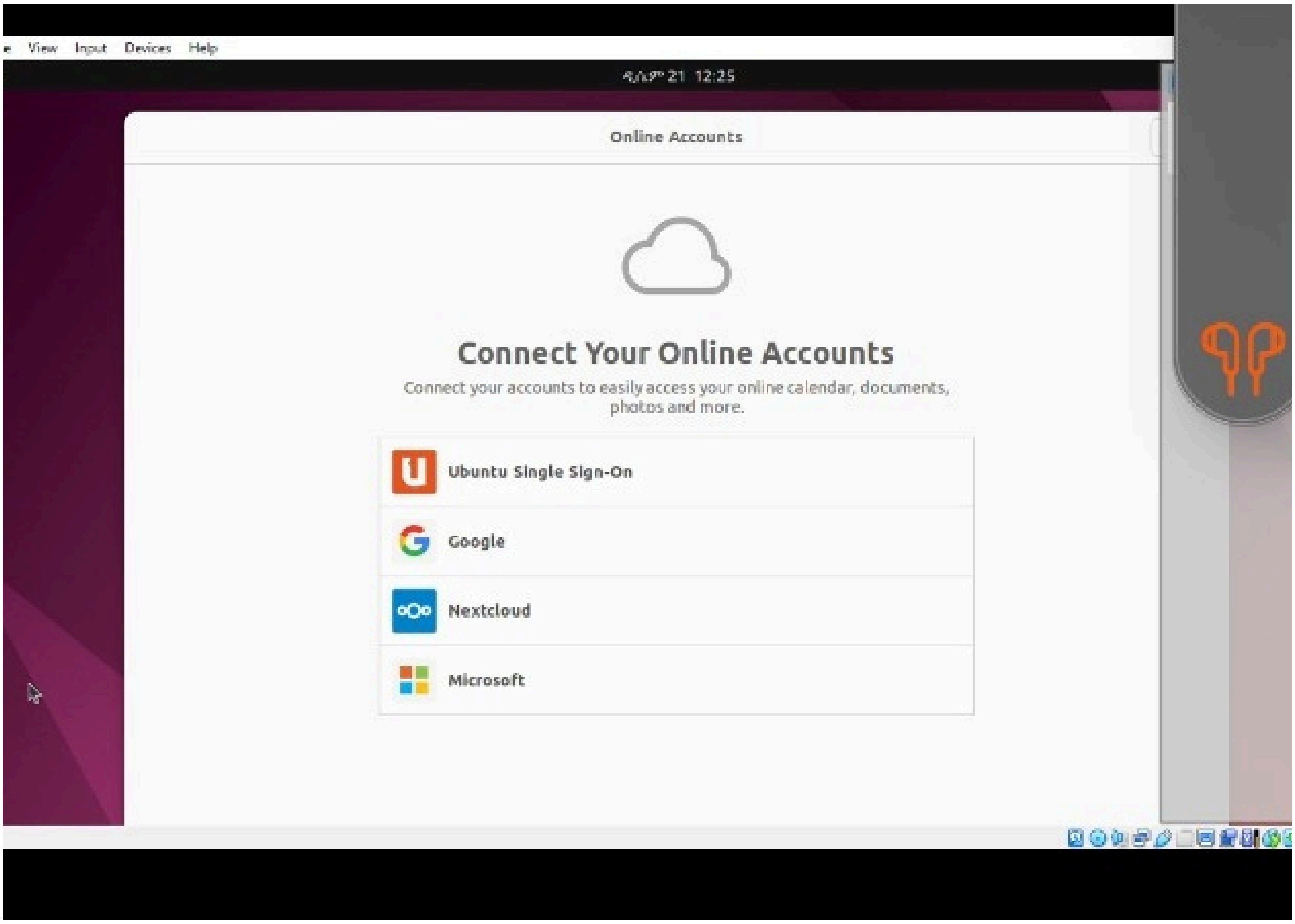

Install
Access for everyone
At the heart of the Ubuntu philosophy is the belief that computing is for everyone. With advanced accessibility tools and options to change language, colour scheme and text size, Ubuntu makes computing easy – whoever and wherever you are.
Customization options
Appearance
Assistive technologies
Language support
Copying files...
Skip
0:17:01
Type here to search
19°C Mostly sunny
ENG
10:01 AM
12/21/2025


Dec 21 10:12
Install
Help and support

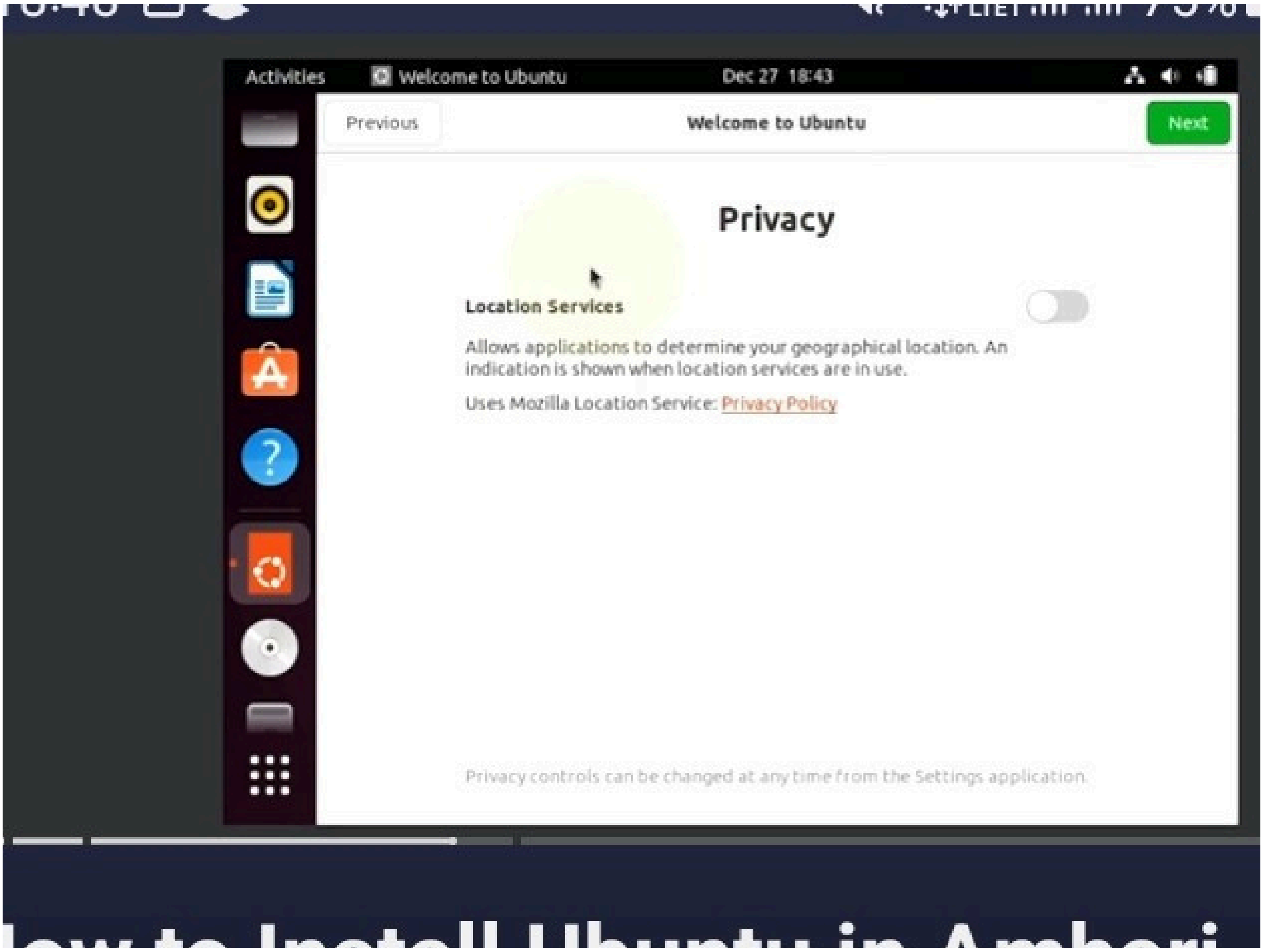

ubuntu 22.04 [Running] - Oracle VirtualBox
Machine View Input Devices Help
Dec 21 10:30
0:23:42
Installation Complete
Installation is complete. You need to restart the computer in order to use the new installation.
Resta Now
Type here to search
21°C Mostly sunny
ENG
10:30 AM
12/21/2025

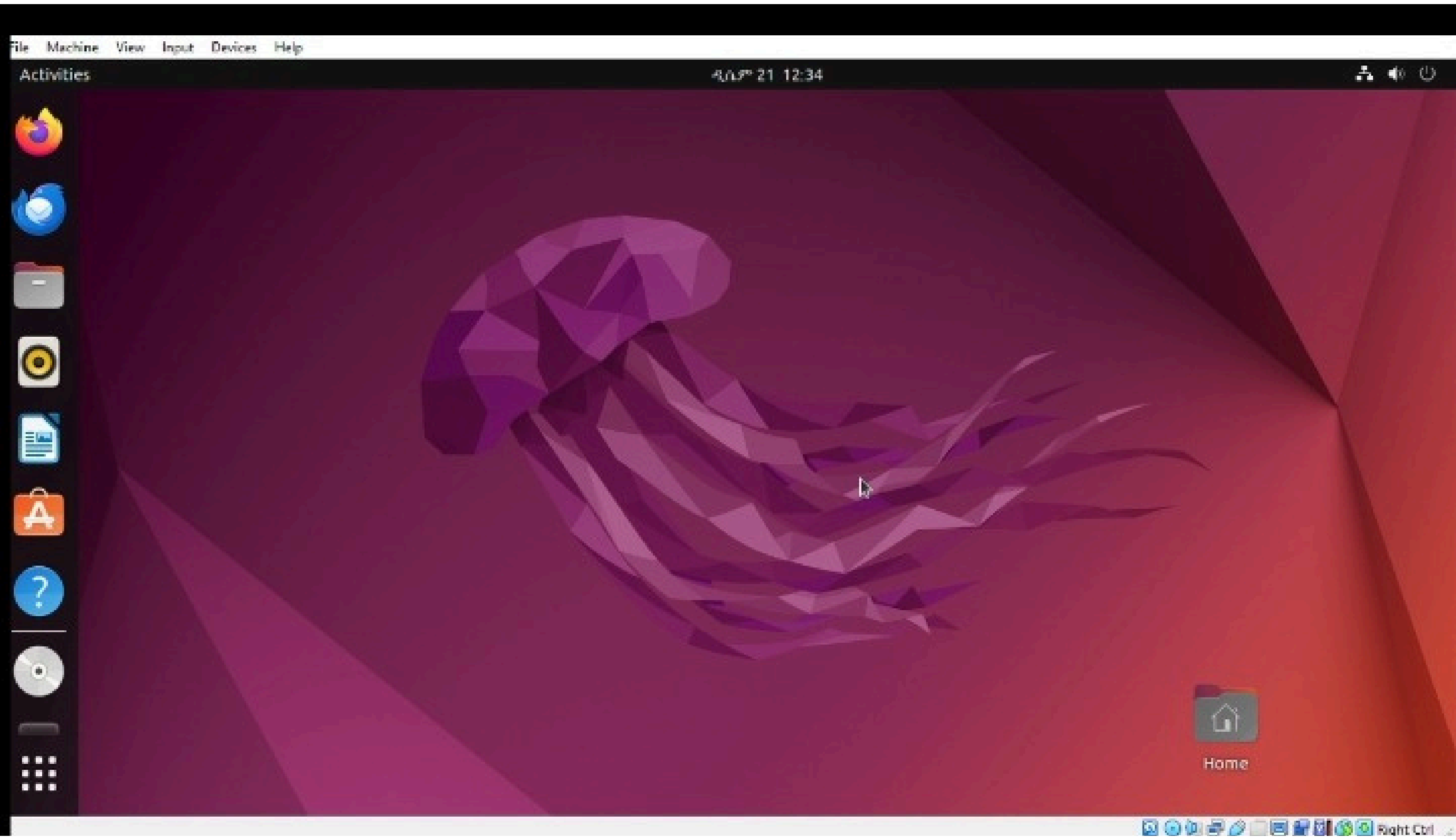

View Input Devices Help
Mouse integration
Please remove the installation medium, then press ENTER:


File Machine View Input Devices Help
21 12:33
habtamu22.04
Ubuntu
Right Ctrl

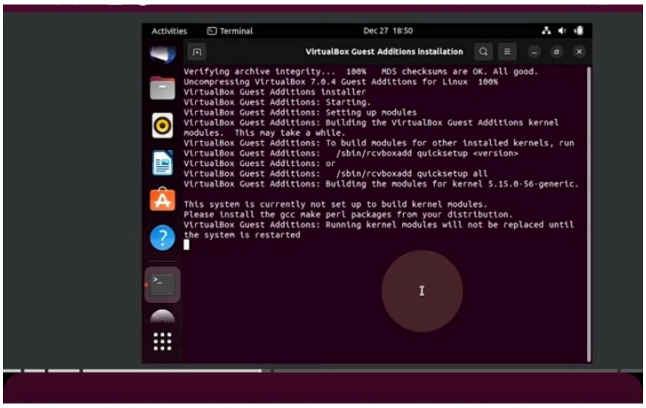

View Input Devices Help
Online Accounts
Connect Your Online Accounts
Connect your accounts to easily access your online calendar, documents, photos and more.
Ubuntu Single Sign-On
Google
Nextcloud
Microsoft


Activities
Welcome to Ubuntu
Dec 27 18:42
Previous
Livepatch
Next
Livepatch
Canonical Livepatch helps keep your computer secure, by applying some updates that would normally require restarting.
Would you like to set up Livepatch now?
Set Up Livepatch...
Legal notice

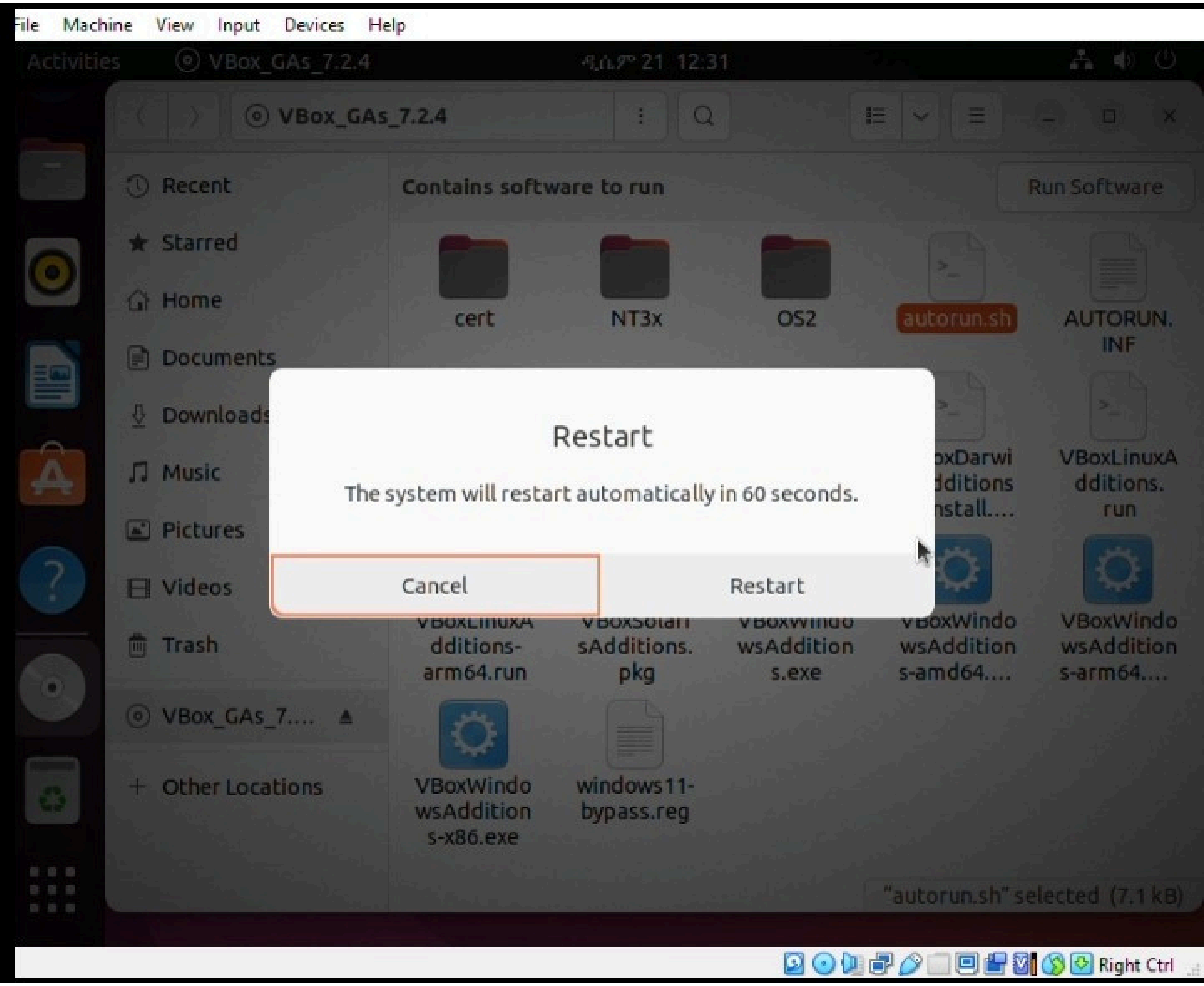

Activities
Welcome to Ubuntu
Dec 27 18:43
Previous
Welcome to Ubuntu
Next
Privacy
Location Services
Allows applications to determine your geographical location. An indication is shown when location services are in use.
Uses Mozilla Location Service: Privacy Policy
Privacy controls can be changed at any time from the Settings application.


File Machine View Input Devices Help
Activities
ዲሴም 21 12:34
Home
Right Ctrl

Activities
VBox_GAs_7.0.4
Dec 27 18:50
VBox_GAs_7.0.4
Recent
Starred
Home
Documents
Downloads
Music
Pictures
Videos
Trash
VBox_GAs_7....
Other Locations
Contains software to run
Run Software
cert
NT3x
OS2
autorun.sh
AUTORUN.INF
runasroot.sh
TRANS.TBL
VBoxDarwinAdditions.pkg
VBoxDarwinAdditionsUninstall....
VBoxLinuxAdditions.run
VBoxSolarisAdditions.pkg
VBoxWindowsAdditions.exe
VBoxWindowsAdditions-amd64....
VBoxWindowsAdditions-x86.exe
windows11-bypass.reg



Activities
Terminal
Dec 27 18:50
VirtualBox Guest Additions installation
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.0.4 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel 5.15.0-56-generic.
This system is currently not set up to build kernel modules.
Please install the gcc make perl packages from your distribution.
VirtualBox Guest Additions: Running kernel modules will not be replaced until the system is restarted



5.Issues Faced During Installation

.Virtualizationn disabled in BIOS

.Black screen after boot

.Network connection issues

.Slowww system performance

6 .solutions

.Enable virtualization in BIOS

.Change display controller settings

.Increase allocated RAM and CPU

.Install Guest Additions or VMware Tools

7. Filesystem Support

Supported Filesystems

.NTFS	.FAT32
.exFAT	.ext4
.Btrfs	.ZFS
.HFS+	.APFS

Best Filesystem for Ubuntu 22.04: ext4

Why ext4 Was Chosen for Installation?

Technical Reasons:

1. Maturity: ext4 has been stable since 2008

2. Performance: Excellent for general-purpose use

3. Reliability: Journaling prevents data corruption

4. Compatibility: Default for most Linux distributions

5. Features:

   · Support for volumes up to 1 exabyte

   · Files up to 16 terabytes

   · Extent-based allocation reduces fragmentation

   · Delayed allocation improves performance

Alternative Consideration: Btrfs

· Advantages: Snapshots, compression, checksums

· Disadvantages: Less mature, potential stability issues

· Decision: ext4 chosen for stability in learning environment

. Optimized for Linux

. Fast and reliable

. Supports large files and partitions

. Low risk of data corruption

8. Advantages and Disadvantages

Advantages of Ubuntu 22.04

. Free and open-source

. Secure and stable

. Strong community support

. Excellent for learning and development

Disadvantages of ubuntu 22.04

. Limited proprietary software support

. Gaming support weaker than Windows

. Learning curve for new users

9.Conclusion

The virtual installation of Ubuntu 22.04 LTS was successfully completed, achieving all project objectives:

Key Achievements:

1. Created and configured a virtual machine with optimal settings

2. Completed Ubuntu installation with proper partitioning

3. Configured network access and user accounts

4. Installed essential software and updates

5. Implemented solutions for encountered issues

6. Documented the entire process comprehensively

Learning Outcomes:

· Practical understanding of OS installation procedures

· Hands-on experience with virtualization technology

· Development of troubleshooting and problem-solving skills

· Enhanced understanding of filesystem concepts

· Preparation for system administration tasks

10.0 FUTURE OUTLOOK AND RECOMMENDATIONS

10.1 Immediate Next Steps

1. Explore Advanced Features:

· Try different desktop environments (KDE, XFCE)

· Experiment with server edition

· Test various filesystem configurations

2. Skill Development:

· Learn command-line administration

· Practice shell scripting

· Study network configuration

10.2 Medium-Term Goals

1. Advanced Virtualization:

· Try Type 1 hypervisors (Proxmox, ESXi)

· Set up virtual networks

· Implement clustering

2. Professional Development:

· Earn Linux certifications (LPIC, Linux+)

· Learn containerization (Docker, Kubernetes)

· Study cloud platforms (AWS, Azure)

10.3 Recommendations for Improvement

1. For Students:

· Practice regularly with different Linux distributions

· Join Linux user groups and forums

· Contribute to open-source projects

2. For Educational Institutions:

· Provide access to better hardware for virtualization

· Include cloud computing in curriculum

· Arrange workshops with industry experts

11.VIRTUALIZATION EXPLANATION

What is Virtualization?

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical computer. Each VM operates as a complete, isolated system with its own operating system and applications, while sharing the underlying physical hardware resources.

Why Virtualization is Important

1. Resource Optimization: Multiple systems on single hardware

2. Cost Reduction: Lower hardware and energy costs

3. Isolation and Security: Separate environments for different purposes

4. Disaster Recovery: Easy backup and restoration

5. Testing and Development: Safe environments for experimentation

6. Legacy System Support: Run old OS on new hardware

How Virtualization Works

1. Hypervisor: Software layer that manages VMs

   · Type 1: Bare-metal (VMware ESXi, Microsoft Hyper-V)

   · Type 2: Hosted (VirtualBox, VMware Workstation)

2. Virtual Machine Components:

   · Virtual CPU, RAM, storage, and network interfaces

   · Guest operating system

   · Applications and services

3. Resource Allocation:

   · CPU time divided among VMs

   · Memory allocated dynamically

   · Storage managed through virtual disks

   · Network traffic routed appropriately

Modern Virtualization Trends

1. Containerization: Lightweight alternative (Docker)

2. Cloud Virtualization: Virtual machines in the cloud

3. Desktop Virtualization: Virtual Desktop Infrastructure (VDI)

4. Network Virtualization: Software-defined networking

THE END